

Trait-based Ecology I

Today's Agenda:

- Quiz
- Project 3: Trait-based ecology
- Trait-based ecology overview

Project 3

- Due April 11th
- Similar to previous projects and can hopefully build upon them
- Inspired by grant proposals
- 2-4 (ish) pages, references don't count against that

Project 3

You'll propose a hypothesis and how you'll test it using trait-based methods

First 1-2 pages: Lay out a hypothesis you propose to test

- Related to things we've covered in class
- Why is it important? Why study it in this system?
- Can re-use old text and references

Last 1-2 pages: Describe how you plan on testing it

- Should involve at least 5 new references relating to traits
- Make sure to clarify what traits are being measured and why
- Write as though this WILL happen. "We will measure"

Project 3

Questions?

Haiku sharing

Does anyone want to share what they wrote?

Trait-based ecology

What is a trait?

- Lots of variation, with different researchers choosing definitions that often are most applicable to their use cases

Trait-based ecology

What is a trait?

I would argue that the best definition is something like:

“A measurable property of something”

Trait-based ecology

Trait observations have 4 critical pieces of information:

Entity: What is the thing being measured (e.g., an individual of a species)

Observation: The context of the measuring (e.g., location and time)

Measurement: The actual measurement itself (e.g., 10 g)

Trait: What is being measured? This should be defined. (e.g., dry mass...)

Trait-based ecology

So, traits might include things like:

- The species range size of *Loxodonta africana* in Africa in 2011, measured as the alpha-hull of all observations that year.
- The mass of an individual of *Numida meleagris* at a given point and time, measured by bagging the individual and weighing them on some scale.
- The leaf mass of a single leaf of a single individual of *Sabal palmetto* measured at a given point and time, taken as fresh weight.
- The length of a particular stomate on a particular leaf of a particular species, measured at some point and time, using a particular method.

Trait-based ecology

So there is a lot of flexibility in the term “trait”!

- Important to be clear about the entity, observation, measurement, and trait
- Lack of clarity can be a big issues
 - *E.g.*, the mass of *Taxodium distichum*
 - Leaf mass? Whole plant mass? Above-ground mass? Mass of all individuals?

Is it a trait?

- Plant growth form (e.g., tree, shrub, etc.)
- Photosynthetic rate
- Maximum lifespan
- Genome size
- Taxonomic Order
- Body size
- Growth rate
- Temperature range occupied
- Molecular sequence data
- Habitat type

Functional traits

Often, scientists will talk about “functional” traits

These are traits that “*impact fitness indirectly via their effects on growth, reproduction and survival, the three components of individual performance*”



Oikos 116: 882–892, 2007

doi: 10.1111/j.2007.0030-1299.15559.x,

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Subject Editor: Pia Mutikainen, Accepted 8 January 2007

Let the concept of trait be functional!

Cyrille Violle, Marie-Laure Navas, Denis Vile, Elena Kazakou, Claire Fortunel,
Irène Hummel and Eric Garnier

Non-functional traits?

Can we think of any traits that aren't functional?

I.e., those that *don't* impact growth, survival, or reproduction?

Why focus on traits?

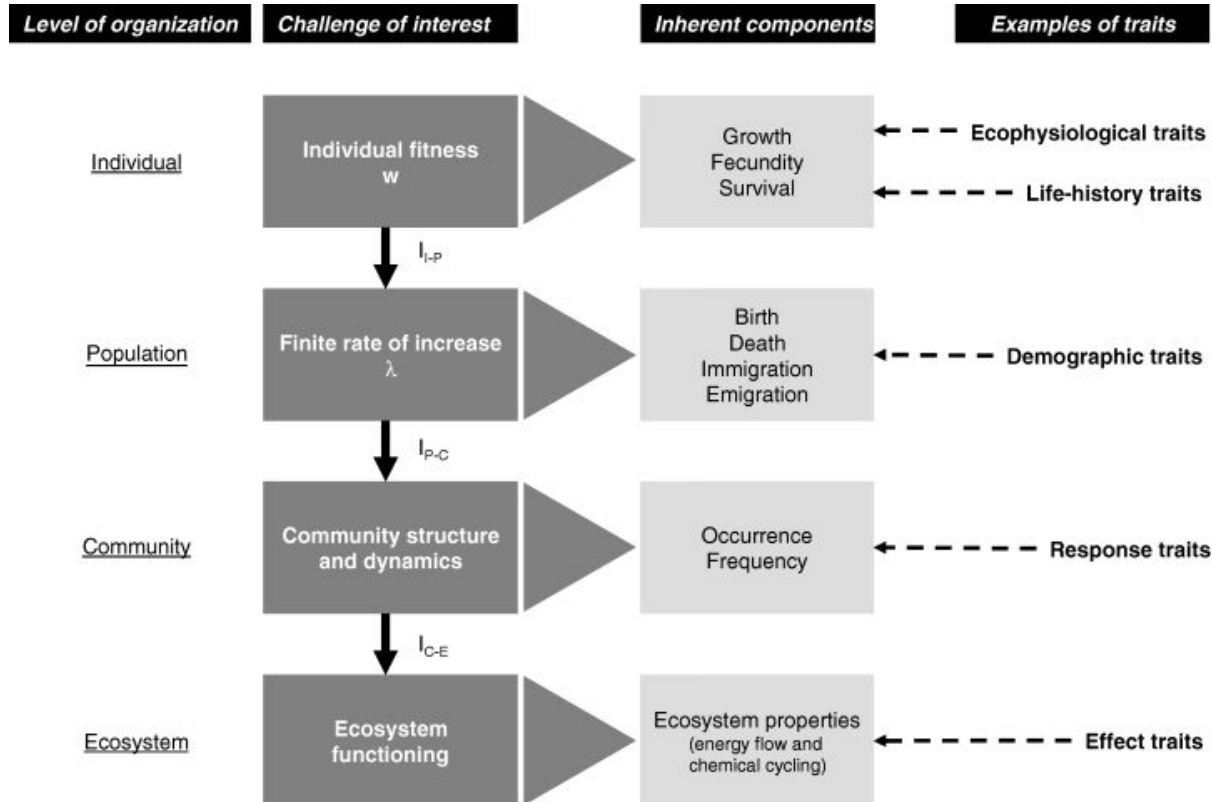
- Generality:
 - Rather than focusing on particular species in particular systems, we can say things about all species in all systems (or at least a larger subset of species/systems).
- Quantitative:
 - We move from species (categorical) to continuous measurements
- Links to mechanism
 - We can focus on traits that are linked to particular mechanisms
 - E.g., if we want to work on predation, mouth size might be useful.
- More predictive
 - Allow us to say something about potential future scenarios



How traits are used

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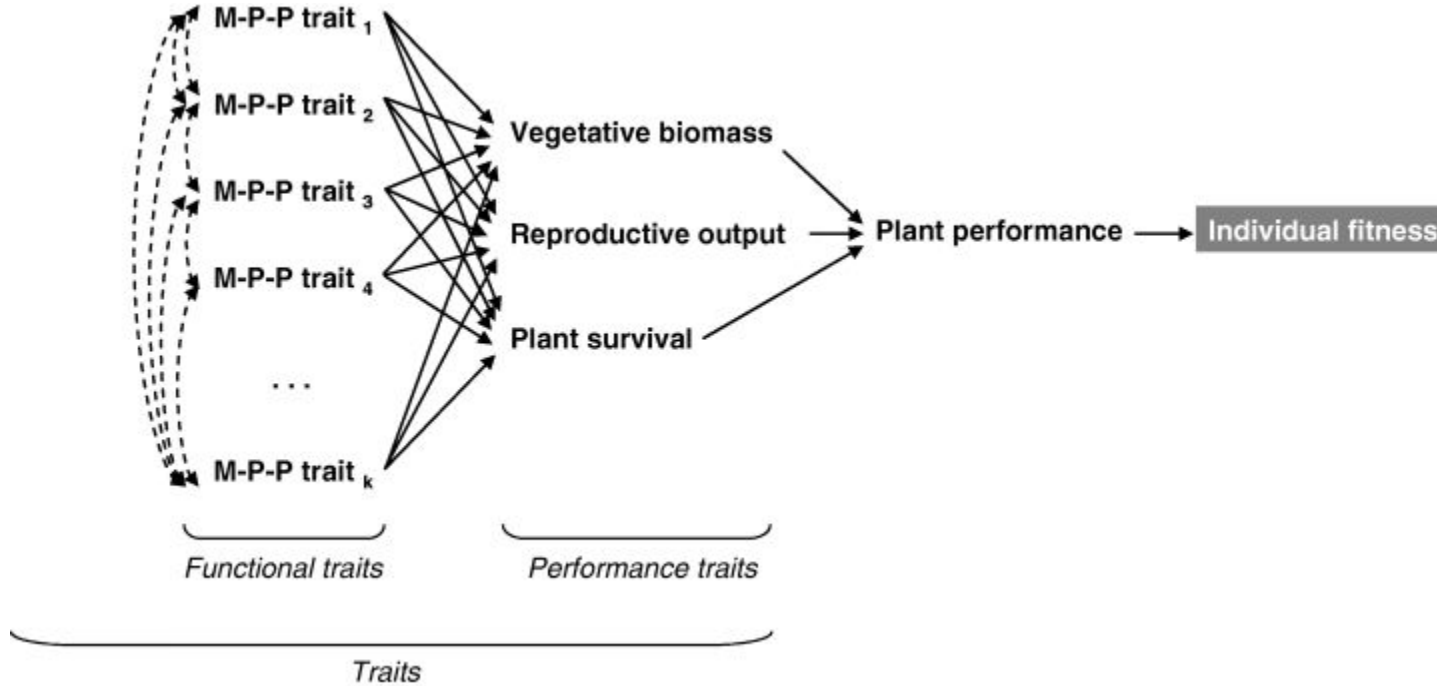




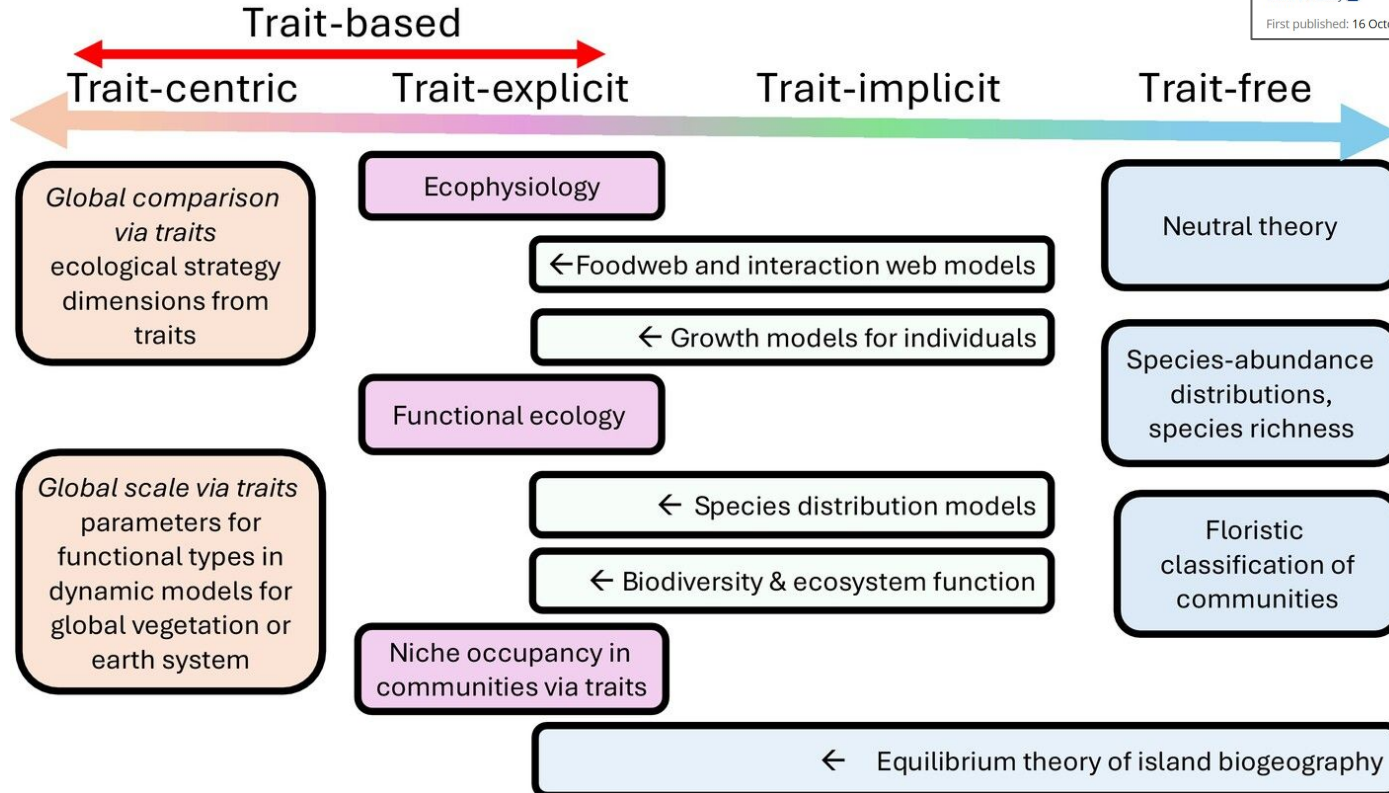
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Trait-based to trait-free spectrum

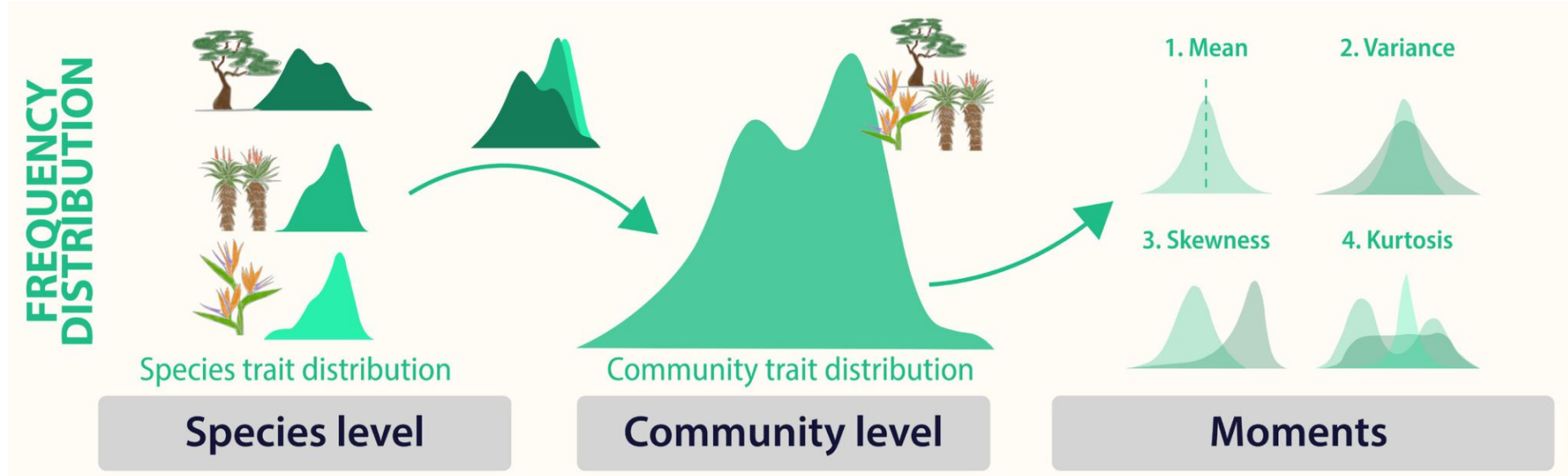


Traits we use

- What traits are people in the class working on?
- Are they functional?
- What are traits are commonly measured in your field? Why?
- Any non-functional traits?

Trait distributions

Traits vary!



Next class: More traits